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20231849

Week 3 Practical

1. VM Deployment Summary

Authentication Justification

Password authentication was selected over SSH key authentication because it is simpler to manage in a learning environment and is appropriate for Windows Server VMs where RDP is the primary connection method. SSH keys are better suited for Linux-based VMs in production environments.

VM Size Justification

The B1s size was unavailable across multiple Azure regions. Standard_B2as_v2 was selected as the most cost-efficient available option, providing 2 vCPUs and 8GB RAM suitable for this learning environment.

2. Network Summary

Network Design Justification

A new VNet (10.1.0.0/16) was created to isolate the VM's network from the Week 2 network infrastructure. A /16 address space was chosen to allow future scalability. The subnet VMSubnet (10.1.1.0/24) provides 251 usable addresses, sufficient for this deployment. A static public IP was assigned to ensure the VM's address does not change between restarts, which is essential for consistent remote access via RDP.

NSG Rules

Rule Name- Allow-RDP

Protocol- TCP

Port-3389

Direction-Inbound

Action- Allow

Justification- Permits Remote Desktop Protocol access to manage the Windows Server VM securely

3. Auto-Shutdown Configuration

Justification

Auto-shutdown was configured at 19:00 SAST daily. In a learning environment, VMs left running overnight consume Azure credits unnecessarily. Auto-shutdown ensures the VM is automatically deallocated outside of working hours, preventing unexpected costs and demonstrating responsible cloud resource management.

The first screenshot shows the 'Net731-VM' overview page in the Microsoft Azure portal. The 'Properties' tab is selected, displaying details for the virtual machine. The 'Networking' tab shows the public IP address 102.133.160.22. The 'Size' tab shows the VM is a Standard B2as v2 with 2 vCPUs, 8 GiB RAM, and 2 threads per core. The 'Source image details' tab shows the source image is 'microsoftwindowsserver' with the plan '2022-datacenter-smalldisk-g2'. The 'Disk' tab shows the OS disk is 'Net731-VM_OsDisk_1_3933ca97fac4b4ea80aedf782f063f' with encryption at host disabled and Azure disk encryption not enabled.

The second screenshot shows the 'Net731-VNet' overview page. The 'Essentials' tab is selected, displaying details for the virtual network. The 'Address space' is 10.0.0.0/16. The 'Subnets' tab shows 2 subnets. The 'DNS servers' tab shows 'Azure provided DNS service'. The 'Virtual network ID' is 412f110f-6e7a-428c-a51a-a0b50c0a0d83.

The third screenshot shows the 'Net731-VM | Auto-shutdown' configuration page. The 'Auto-shutdown' toggle is enabled. The 'Scheduled shutdown' is set to 7:00:00 PM. The 'Time zone' is set to (UTC+02:00) Harare, Pretoria. The 'Send notification before auto-shutdown?' toggle is set to Yes. The 'Webhook URL' and 'Email address' fields are both empty.

